

☰ Hide menu

What is an Integer Linear Program?

NP-Hardness of Integer Linear Programs

Tutorial: Solving Integer Linear Programs using Python/Pulp

Vertex Cover as Integer Linear Program

✔ Video:

Vertex Cover as an Integer Linear Program

6 min

📖 Lab:

Interactive Notes: Vertex Cover as an ILP

1h

📝 Graded Assignment:

Vertex Cover Problem ILP formulation

3h

Linear Programming Approximations of Vertex Cover

Solving Integer Linear Programs using Branch and Bound Algorithm

Problem Set # 2

Vertex Cover Problem ILP formulation

100%

100% Highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

Review Learning Objectives

Instructions

1. Which of the following statements is true about an optimal solution to the vertex cover problem?

1 / 1 point

●

It includes the minimum number of vertices that cover all edges

○

It includes all vertices of the graph

○

It includes all edges of the graph

○

It includes the maximum number of vertices that cover all edges

✔ Correct

Correct! An optimal solution includes the minimum number of vertices that cover all edges.

View submission

See feedback

Try again

2. Which of the following is a correct integer linear programming formulation for the vertex cover problem?

1 / 1 point

○

Minimize $\sum x_i$ subject to $x_i + x_j \leq 1$ for every edge (i, j) and $x_i \in \{0, 1\}$ for every vertex i

○

Maximize $\sum x_i$ subject to $x_i + x_j \leq 1$ for every edge (i, j) and $x_i \in \{0, 1\}$ for every vertex i

○

Maximize $\sum x_i$ subject to $x_i + x_j \geq 1$ for every edge (i, j) and $x_i \in \{0, 1\}$ for every vertex i

●

Minimize $\sum x_i$ subject to $x_i + x_j \geq 1$ for every edge (i, j) and $x_i \in \{0, 1\}$ for every vertex i

✔ Correct

Correct! In vertex cover problem, we aim to cover all edges with minimum number of vertices.

3. If you add a constraint ' $x_i + x_j \leq 1$ for some edge (i, j) ' to the vertex cover problem, what will be the effect?

3 / 3 points

○

The problem becomes infeasible

○

All valid vertex covers will be excluded

○

The problem remains the same

●

Some valid vertex covers will be excluded

✔ Correct

Correct! Adding the constraint ' $x_i + x_j \leq 1$ ' excludes the vertex covers that include both vertices i and j .

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